Assignment-II

EE655: Computer Vision & Deep Learning

Q1) Use the shared blink detection dataset to train a conditional GAN for generating synthetic images of open and closed eyes. Generate and provide at least 10 images for each category. Clearly describe the conditioning mechanism used. **[10 marks]**

Q2) Develop a tool to count the number of blinks of a patient in the given video. Use a pre-trained facial landmark detection model just to detect and crop the eyes appropriately, and pass these cropped eye images to a classifier trained on the shared blink detection dataset. Describe your classifier and report blink counts separately for the right and left eyes. **[10 marks]**

Q3) Cervical Dystonia is a painful condition affecting the neck, causing abnormal head tilting, as illustrated below. Using a pre-trained body landmark detection model, design appropriate features to identify such head poses. Describe your features clearly (with diagrams) rationally and report their values for both normal and such abnormal poses. **[10 marks]**

Head turns to side
(rotational torticollis)

Head tilts to the side
(laterocollis)